

Lec 28 Latent Space Interpolation

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Transcript

[00:00] Hello everyone. Welcome to the lecture series on fundamentals of generative AI and the large language models. We are in the week four and the week four is all about the advanced VAE concepts and the practical applications. And in the in this particular session, let us look into one of the most important concept called latent space interpolation. And uh by now you must have understood that VAE is mainly responsible for generating and uh data.

[00:31] If it has to generate the data, what are all different ways? One of the important concept has to be an interpolation. What is interpolation? Let us first understand and then we will get into the math behind. So, this is what is an interpolation. Now, whenever you want to do an interpolation, you require two inputs. You require input A and B. So, here let us take the start image or the first image as a three and the end image as eight. Now, uh so far from our previous session and from all the sessions of VAE, we have understood that whenever an input is being passed to an encoder, encoder has to give us the latent space.

[01:17] And the latent space cannot be a single number. It will be a vector. All that we have already seen it. So, now let us understand that when an input three is

a passed to an encoder, encoder has obtained or given this particular latent space. Okay. What is the next step? Now, by now you should be knowing that how to obtain the latent space, right? In case if this is not very clear, you can go back to the previous week, week three and the last session we had done a numerical example. So, you can once again look into that particular session before you come here. So, that you will be able to understand how each of these you know terms you will be understanding more clearly. Mathematically also you

[02:04] will be understanding. » [snorts] » So, input is passed to an encoder and we obtain the latent space. Once you get the latent space, what is that we are passing that latent space to a decoder? What is the role of a decoder? Decoder has to generate the data back or reconstruct the data. So, now [snorts] when you have passed this input to an encoder with this latent space when you passed it to a decoder, let us assume that we obtained this as a reconstructed image. Although it will not be a blue, it is just for a visualization. So, you will be having a reconstructed of this particular three. Not exactly three, you will be having an image which looks similar to three. Maybe little blur. So, but you will be getting three

[02:54] but not this blue box. No, this is your reconstructed image. Now, you take this input. Forget about the rest middle three, whatever. I'll come back to that. Now, we have an end image. What is this end image? Here the image is hand digit eight. That you pass it to an encoder. What is the role of an encoder? It has to produce us the you know latent space. This latent space pass it to an decoder. What is the role of a decoder? Decoder has to reconstruct or generate the data back. So, here let us assume the decoder has given this particular data. Now, what is interpolation? Interpolation here is now you want intermediate data, in between data. Now, if you have in between data, all what you have is input. This input

[03:50] and this input is all what we have. But, from this input, we got the latent representation. Is it not? We got this as the first latent representation, and we got this as second latent representation. Is it not? We definitely have these two with us. Yes. Now, once we have these two, which means here you have in terms of a number. It is a vector. It is in terms of a number. Here also you have in terms of a number vector.

[04:20] Right? Now, from both this latent space, can I do some sort of a mixing? Can I do some mathematical operation so that from this can I get this Z? Right? Now, from this can I get this Z? From this can I get this Z? Now, at the same time, you cannot use only one latent space. I cannot use only this and then get this. Only this and then get this. Because what we are discussing is interpolation. Interpolation is between these two.

[04:58] Can I use some sort of a technique or math where you have this latent representation, and you have this latent representation? Can I combine these two latent

representation very systematically, and can I get this? From these two, by combining in a systematic way, can I get this latent representation? Can I get this latent representation? So, from this and this, when you are getting these three, is it not the from there when you pass it to a decoder, decoder is going to reconstruct or generate the data. And that data will be definitely similar to between three and eight only. Are you getting this intuition? So, now our ultimate aim is you want the intermediate images.

[05:53] But, all what we have is the first image and then the last image. We have only these two. From these two, if you have to get the middle images or intermediate images, but we cannot imagine what can be those intermediate. So, between three and eight, not necessarily that it can you know it can be any you know image that can be between three and eight. But, systematically if you have to interpolate between these two, then all what we have is the latent representation of three and latent representation of eight. So, if I have a latent representation of these two, and so, which means I have this latent representation and this latent representation.

[06:40] Some systematic way, can I combine these two latent representation and get the intermediate latent representations? Can we think in that way? It's possible, right? Once I have one vector and other vector, can I use some means to combine these two? How to combine we'll see in the upcoming slides. But, by combining, is it not I get one more latent representation, one more Z I get? By combining in a different way, you get one more Z. By combining in one more, you get one more Z. So, once you get a different Z, each Z is passed to a decoder. Role of a decoder is to reconstruct. Is it not this Z will be very much similar to this and this? This Z will be very much similar to these two.

[07:28] Not very much, but it will be closer. If that is the case, intermediate images also will be closer. I hope the intuition is a clear. I hope the theoretical concept is a clear to you. This is all about the interpolation. Now, you can ask where exactly we will be using this. One of the classical example is say I have we are working on medical application. We have a patient who has come when the patient was in an initial stage of some tumor.

[08:00] Then, we have you know, the patient did not come back to hospital again. Patient went back to some other hospital. Then, after well grown tumor, the patient came back to the same hospital. So, in between we don't have the data at all. So, but ideally we want to analyze how the tumor has progressed over a period of time. How did what was the size? So, in such application, the patient did not come back to the hospital, but we want to see how the progress would have been. So, in that case, I have an initial image, I have an end image. So, with that, can I get an intermediate images? Will it not be much beneficial for a analysis, better diagnosis? Can we think can you imagine can you imagine this

[08:47] particular application? So, this is where we have an interpolation concept

which comes into. And how is that we do? We can do that with very much with VAE only, but all what we need is an interpolation concept. Which is a simple math. Let us see in detail how it is. First, you have passed an input image three. So, let us assume those features image which is in matrix form. When you convert into convert it into a single vector, is it not you have one dimensional one? That you pass it to an hidden layer. See, this is exactly similar to week three last numerical example. You can just go back once so that you will understand what exactly is being explained here. So, since the latent representation size is two, we

[09:36] get a two mean two log variance. From log variance, you'll be converting it into an a variance and then the standard deviation. We we know that. So, we are getting a variance. Since the latent space is a two dimension is two, we get a two mean two variance. From this, can I get a Z? So, what is that? Is it not we get Z1 Z2? So, instead of Z1 Z2, here I'm slightly changing the names. Instead of calling it as Z1 Z2, I'm calling it as Z A1 A2. Why Z A1 A2?

[10:10] Because this is the first image. That first image we are considering it as XA. I hope this is clear to you. So, this is how you calculate Z A1 as mean one plus a variance sigma, and then you have this particular uh value sampled from a normal distribution. Right? You do this. Now, you have even Z2 also, Z A2. Now, let us assume this is the mean value that we have obtained. And from towards the end, we have calculated, and this is the Z A value that we have obtained. When I say Z A, it is Z A1 Z A2. Or from that particular session terms, if you have to write Z equal to Z1 Z2. This is what we have obtained. I hope this is clear so far. Nothing we have done. We have passed an input to two an encoder. Towards the end of the encoder,

[11:07] we are getting Z1 Z2, Z A1 Z A2 for this particular input. Which means what we have discussed so far, we have we have reached to this step. This is what we have we have reached to this. Okay. Now, let's go back and then see what will happen when you pass the second input. Is it not the steps will remain the same? You have this input, which means you have to pass it into an encoder. And the encoder will give these parameters. From this, is it not you will get a Z? So, you will get now ZB.

[11:41] Why ZB? Because this is a second input, which is denoted as XB. And then since the latent representation is having a two vectors, ZB1 and ZB2. Right? This is how you calculate it. I hope up till this part it is clear to all of us. So, let us assume this is a ZB latent representation that we have obtained. How did we get? I'm just giving one a direct example. After doing all of the calculation, let us assume this is ZB that we have obtained. In order to understand, you can solve the problem once again, so that you'll be able to understand this. Okay. Now, what we have with us is this. We have ZA and we have ZB. Which means go back to our initial diagram. We have an input that from that input we got the latent

[12:31] representation. We have a second input, from there we got the latent repre-

sensation. That is what are ZA and ZB. How did we get? It is a simple feedforward calculation. Now, from here, now you need an intermediate. Let me use another pen, so that you'll be able to understand. Now, you need an intermediate latent one. Can we have an intermediate latent so that when this is a passed to an decoder, we get one image. When this is a passed, we get one image. When this is passed, we get one image.

[13:09] One image. One image. And this image and this image intermediate will be closer. That's the whole concept. Okay. So, which means now you require that intermediate latent representation. So, now we perform latent space interpolation. Why? Because input and output you cannot perform interpolation. All what we have is the compressed representation of that input, which is latent space. How is that you do? You require some systematic way to combine those two latent space. So, this one and this one you are considering and from here in a systematic way you are combining those two latent space. What is the formula we are using? It is $1 - \alpha Z_A + \alpha Z_B$. Now, what is alpha here? It is an

[14:07] you know parameter hyper parameter or interpolation parameter that mainly controls the movement. How much you want to move from ZA to ZB? How much So, in a very systematic way you are using you are moving from ZA to ZB. So, so that you will have when you are moving from ZA to ZB, is it not we are getting the intermediate representations? Right? This is all about the interpolation. So, now how are we doing? We are taking both these latent representations and we are combining.

[14:41] How are we combining? This is the formula that we have used. $1 - \alpha Z_A + \alpha Z_B$. Now, what is these alpha values? Alpha values are those intermediate values. You see, if the alpha value is zero, suppose if you take alpha value as zero, then substitute the values in this. Can you substitute zero? Zero into any term, then $1 - 0$, you are getting Z_A . If it is a Z_A , is it not this latent representation? This Z_A latent representation when you pass it to a decoder, you get this original input to one only. That is what when alpha is a zero, we get exactly Z_A . How Z_A ? Just substitute the values back into that formula. And since zero into something will be zero, $1 - 0$ is 1. 1 into

[15:37] Z_A is Z_A . We get exactly Z_A , which actually represents the image three handwritten image A or image three. Similar way, what is the last value? The maximum alpha value that you can take is between zero and one. Between zero and one only we can take. We have to take. Now, what is the alpha value? Alpha value if it is one, what is going to happen? Substitute $1 - 1$ value is zero. Zero into something this term becomes a zero. 1 into Z_B is Z_B , right? We get exactly Z_B . Which is nothing but this and when you pass that to a decoder, we get exactly that image B. What is image B here? Handwritten eight, digit eight. So, you you understood this concept here.

[16:28] Alpha value can be anywhere between zero and one. When you take it as a zero, we get the first image. When you take it as one, we get the last image. That is the second image. Uh now, in between we need. So, what should be the alpha value? So, here in between alpha values what we have considered uh are 0.25, 0.5, 0.75. These are the in between values that we have considered. Now, let us to calculate uh uh how exactly we get when alpha value is a 0.25.

[17:02] » [snorts] » Now, uh you can look into this particular formula. Here, this is uh since our latent space, uh for example, here is having a two dimension. There are two such values in our latent space. So, if that is the case, how is that you expand the formula? The formula is given here. So, it is once again the same, but since there are two terms, we are using $Z A_1$, $Z A_2$, $Z B_1$, $Z B_2$. But, look into this. This entire is with respect to A one. Okay, this entire is with respect to second. So, this is the rather than me reading out this formula, you can see so that you will be able to understand it much better. So, this is the overall formula, but the since we have two such terms, we are having two here.

[18:02] Ultimately, we should get the two, right? What are those two? Those two are first one is with one, that is $Z A_1 B_1$, and the second is with A_2 and B_2 . Let's take with one numerical example here. You will be able to understand. Now, what is you know, this is $Z A_1$. This is $Z A_2$. This is $Z B_1$, and this is a $Z B_2$. Right? This is what are the values. And how did we get those values? These values are these. Now, perform a simple calculation. What calculation? Whatever is the formula that is given here. So, what is the formula?

[18:47] So, let me erase this so that it will be clear to you. So, the formula here it is $Z A_1$, $Z A_2$, this is $Z B_1$, this is $Z B_2$. Now, substitute the values into this formula. What is this? $1 - \alpha$. What is alpha term? Alpha term is 0.25. $1 - 0.25$, is it not 0.75? Right? So, that is what is 0.75. Into $Z A_1$. What is $Z A_1$? This is $Z A_1$. You have written. Plus plus you have this plus. This is alpha. What is alpha? 0.25. 0.25. What is this?

[19:31] $Z B_1$. What is B_1 ? This is B_1 . You have substituted. I hope you have understood. Perform the calculation, you get the first term. Now, you should get the second term. Why you should get second term? Because see $Z A$ is a two-dimensional one. If it is two-dimensional, intermediate also should be two-dimensional. This also should be two-dimensional. And $Z B$ is once again two-dimensional. So, all our intermediates should get two values because there are two here. If you have to get the two, first we have calculated. Second term is this. $1 - \alpha$. Since alpha is 0.25, $1 - \alpha$ is 0.75.

[20:17] This value. And what is this? $Z A_2$. What is $Z A_2$? This is $Z A_2$. Plus alpha. Alpha is 0.25. And it is $Z B_2$. This is $Z B_2$. This is $Z B_2$. Calculate and you get this value. Now, understand what we have done. All what we had was a $Z A$. We had $Z B$.

From $Z_A Z_B$, is it not we have obtained intermediate latent representation? Is it not we have obtained? And what is this? It is 0.25. What value we have obtained? Now, can you see this value and what value we have obtained in Z_A and Z_B ?

[21:03] Is it not slightly it is a matching? Can you see? It is very much a matching closer to this. Closer, I'm not saying exactly, but it is a closer. Which means, now, if Z_A is a three, the next one will be very much closer to three only, but not exactly three. So, look at this Z_A and look at this value. Is it not the value Look at the second term. Is it not a little high and first one is little less? So, is it not when this value is passed to a decoder, decoder will generate an image very much closer to three?

[21:42] Are you getting this? Right? Yeah. Now, similar way, alpha value is 0.5. So, same calculation once again you perform. Since alpha value is 0.5, $1 - 0.5$ is 0.5. And alpha as it is 0.5. So, get the values from our previous slide. From here, get $Z_A Z_B$ value. Substitute in the formula given and then get the intermediate latent representation. This is what is the intermediate latent representation. Then, take alpha as 0.75. $1 - 0.75$, we get it as 0.25. That is what is 0.25.

[22:29] And alpha value as it is. Okay? So, now, are you seeing In the previous one, here you are taking 0.75 portion of Z_A , 0.25 portion of Z_B . Right? Then, here you are taking 50% of Z_A , 50% of Z_B . Then, here you are taking 25% of Z_A , 75% of Z_B . So, therefore, you have obtained the values in this way. I hope this is clear to you. You know, what we have done is we have obtained an interpolation. And interpolation on what? Not on input, but on latent space.

[23:13] Finally, is it not this is the value that we have obtained? So, Z_A when you pass it to a decoder, ultimately it is the latent space which is denoted as a Z that has to be passed to to a decoder. When decoder when you pass it to a decoder, decoder will give us the reconstructed one or generated one. So, when Z_A is a passed, you get X_A hat. When Z 0.25 latent space is representation is passed to a decoder, we get this value.

[23:45] When Z 0.5, which is this value. When you pass it to decoder, you get this. When 0.75, this value. When you pass it to decoder, we get this. Finally, Z_B when you pass it to decoder, we get this. What did we do? Did you get this now? We have come to an end of an interpolation. So, what is that we have achieved? Our ultimate aim is it is a generation concept. If it is a generation generated one, it has to generate some data. So, if it has to generate the data, I have only input and this input. First input and second input. With the inputs, I cannot get intermediate inputs images. But from input, you have converted into a latent space. That latent space if you can interpolate it, that if you pass it to a decoder,

[24:32] decoder will ultimately generate the data. Which will not be similar to this X and this you know, three and eight eight. Is it not very slowly you are moving from three to eight? Right? How much Now, you're not moving in a random way.

You're going in a systematic way. You're taking some portion of this, some portion of this and you're combining. Next time you're taking some more portion of this, some more portion of that and you are combining it. So, finally the generated images will change something like this. First, you have an digit three, then you will get the three like intermediate image, then you get a mixed of three and eight image, then you get eight like an image and you get a digit eight. So, this is all about the

[25:18] interest latent space interpolation because we are moving smoothly systematically between the two latent points in a very learned latent space. So, not in a random way, in a very systematic way we are moving. So, if you have to understand this diagrammatically, so can you able to see this? You have an input image here and you have another input image and can you able to see this part how it is getting changed? I want you to visualize this.

[25:56] So, for a better understanding, I've taken it from an internet source. I give the complete credit to this particular uh author. So, is it not as you are changing, you are getting an different variation of and towards the end is it not moving closer to that particular second image? So, this is what is an latent space interpolation. One of the important applications, especially whenever you are working on those applications where you don't have an intermediate data, but you require to analyze that intermediate data also, then latent space interpolation will be very good concept which you can explore.

[26:40] With this, we have come to an end of week four. And in the next two sessions, you'll be seeing an hands-on implementation of the concepts that you have learned in this particular week. Thank you. »